

What Does Your

Signature Mean?



Ever wondered what those cryptic characters that some geeks use as their signatures, mean? I like to call them command-line gizmos.

I hang out a lot on *usenet*, mostly on *comp.unix.shell*, a group that has educated thousands of shell users for many years. Many of the posters here have signatures that are small scripts. Deciphering these scripts is very entertaining and pushes you to the man pages.

So I also started sporting these signatures. One good thing is people stop by and ask me about that cryptic text I sport at the end. When I explain, they get quite pepped up about UNIX/Linux.

My current signature is:

```
export BC_LINE_LENGTH=10000
awk 'BEGIN { for (i = 1; i <= 999; ++i) print i } | paste -s
-d "*" | bc |
perl -lpe '%/*?0+$/length($1)/e'
```

So what does this script mean? Let me explain. An interviewer once asked me the following question.

Q. $x = 1^2 + 2^3 + \dots + 999$

How many trailing zeroes will you get in the resulting number, x ?

And being groomed in the UNIX way of life, I wrote the script I've mentioned above, and told him we can find out the number of zeroes *this* way...

He liked it (though this was not an expected answer) and moved on.

When I came back home, I tried the same script out for a much larger range, up to 9999, liked it and made it my siggie. I also realised I was using some common UNIX command line idioms.



Note: For newbies, all the above commands in the 'pipeline' are 'filters'—programs that read from standard input, do something useful and write it to standard output. You can run all these filters separately—as *simple commands*—so you can experiment with sample keyboard input and screen output. Filters don't care how they are used. It is the shell that arranges things; so filters, *unknowingly*, become part of a pipeline and without knowing it, talk to each other.

Going back to my signature – let's look at the second line:

```
awk 'BEGIN { for (i = 1; i <= 999; ++i) print i } | paste -s
-d "*" | bc |
```

In *awk*, *BEGIN* block executes as a preprocessing step... before any input is